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Gaming for Sustainable Futures: Exploring the Role of Environmental Empathy in Worldmaking Interactions

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***Abstract*—It has been suggested that we currently experience a crisis of the imagination, which render us unable to envision and embrace futures radically different from the present. Given the ongoing climate crisis we are in dire need of new, sustainable, visions. Researchers have claimed that such futures can be co-produced by incorporating principles of Worldmaking Interactions in interactive media. In this paper, we explore this idea specifically within the realm of video games, and further develop the concept of Worldmaking Interactions within this context by incorporating recent insights from research that suggest that Environmental Empathy is crucial for fostering sustainable interactions with the biosphere. The empirical results reveal that players often exhibit empathetic responses towards in-game content that can be translated into actions and concerns related to real-world issues.**

INTRODUCTION

The pervasiveness of video games has recently reached unprecendeted levels, especially in Western societies. The majority of the inhabitants in European countries engage in various forms of digital gaming. While it is safe to assume that most people engage with video games due to the enjoyment it provides,[1] researchers and practitioners have in the past couple of decades started to reflect on whether it is possible to make use of the medium as a vessel for other productive purposes, such as educating players about various real-life issues. Under the umbrella of *gamification*, designers have sought to integrate game elements like badges, leaderboards, achievements, and points into non-game contexts to boost motivation and participation. A related phenomenon is *serious* or *educational games* that are games designed with the explicit purpose of impating knowledge or facilitate learning. For example, *Foldit* is an online puzzle game that invites people to contribute to scientific research by solving complex protein-folding puzzles. Furthermore, conventional video games can contain modules or game modes inspired by serious games. Among others, *Assassin's Creed Origin* comes with a a "discovery tour" game mode where the player can walk around ancient Egypt as if it was a virtual museum, and *Europa Universalis* explicitly strive to educate players about real-life history, politics, technology, and military strategy.

It is well known that many video games, especially those targeted for a mainstream audience, are designed to cater to power fantasies where the player is encouraged to either strip the in-game world of its resources and reach world domination, or defeat external threats by brute, lethal force and questionable ethical decisions. Despite this, the last decade has seen an increase in using game design to raise awareness of sustainability-related problems and issues.[2] This is sometimes contributed to the medium's potential to promote sustainable practices in light of global scarcity scenarios.[3] For *SimCityEDU: Pollution Challenge!*, which is a special edition of the famous city builder *SimCity,* the aim is to educate middle-school students about environmental dynamics in real cities, and in *Civilization VI: Gathering Storm*, players are encouraged to address resource-related challenges in a world plagued by extreme weather conditions caused by

industrial activities. Many survival games, such as *Fallout IV* – often set in dystopian futures affected by conflicts or climate-induced disasters – can help to further players' understanding of resource-related challenges in a future world of scarcity.

While we are largely sympathetic towards the increased popularity of sustainability-related themes in video games, we remain critical of how these themes are conventionally presented. The focus on resource management can be deemed reasonable since we expect a shortage of certain critical materials in the near future.[3] Unless we alter current unsustainable trajectories, an environmental collapse is inevitable, a conclusion reached already in 1972 in the now iconic report *The Limits to Growth* by the *Club of Rome.* Still, as articulated by Chang and others, video games often portray their environments merely as collections of usable resources.[5,6] The player is typically depicted as someone who exploits and exerts control over these environments and their inhabitants to amass wealth and power within the game, and the interaction with the environment is limited to relatively straightforward actions, such as building, hunting or foraging. In most roleplaying games (RPGs) the vast population of non-playable characters (NPCs) inhabiting the land exists exclusively to be hunted for quests, loot and various other usable resources. A too narrow focus on resource management risks boiling sustainability down to a "complex problem to solve" within existing systems and paradigms.[4] Heijmen and Vervoort emphasize the need to redirect human experiences towards a more interconnected relationship with nature, and to produce experiences of embeddedness to decrease the likelihood of environmental harm.[6] Therefore, Chang suggests that games need to be designed to more accurately depict our entangled relation between players and their surroundings.[5]

Drawing on such insights, this paper aims to show how conventional games can be used to further environmental awareness and action towards sustainability-related purposes through drawing on a design theory from Human-Computer Interaction (HCI) called Worldmaking Interactions (WI). Developed by Roy Bendor, the notion of WI goes beyond mere *worldbuilding* – i.e., designing realistic or probable scenarios, environments and characters in-game – and focuses on how interacting with interactive media can provide opportunities to collectively visualize and construct new and sustainable futures. We see that while climate change and other environmental effects and disasters resulting from human, industrial activities will inevetably require us to rethink resource management (through for example a Circular Economy), sustainability should not solely be seen as a "tool" to avoid a climate catastrophy, but as a different way of relating to the world. A sustainable future is thus not necessarily a clean and green version of the present, but sustainability can instead be seen as a plethora of more or less desirable futures that need to be collectively visualized and strived for through finding and engaging with trajectories pointing towards those futures.[5,6]

We have three primary objectives with this paper. First, we aim to translate the concept of WI into a video game design context to enhance "the public's own ability to imagine alternative futures" through these interactions.[4] Second, we aim to provide empirical examples of how existing video games afford such interactions in-game, and how they translate into real-world contexts. Third, we aim to promote the emerging concept of Environmental Empathy (EE) as a component to WI to ensure an explicit focus on environmental sustainability in discussions about and visions of desirable futures.[7]

THE CRISIS OF THE IMAGINATION

It has been argued that one key obstacle for sustainable transitions is that we lack clear and positive visions of possible futures.[8] Many thus claim that the environmental crises we face today are the result of a crisis of the imagination. This refers to the challenge of imagining futures that are radically different from the present – especially during times like ours when change is necessary, urgent, and inevitable – since we are entrenched in existing paradigms, norms, or systems.[9,10] This poses a challenge for radical innovation, social progress and transformative change. Instead, we mainly explore sustainable transitions within the existing paradigm of an unsustainable status quo, which has led to an almost exclusive focus on technological development and energy and resource efficient processes.[11]

Haiven discusses the pervasive influence of capitalism in the collective consciousness, emphasizing the political dimension of the crisis of imagination, often perpetuated by political interests.[9] Margaret Thatcher's now famous slogan "There is no alternative" exemplifies this restrictive mindset and its potentially harmful consequences. Mark Fisher has coined the term *capitalist realism* to describe the prevailing belief of capitalism as the only viable economic system, which severely limits the potential to conceive credible alternative futures, for example one without

continued economic and industrial expansion.[12] The crisis of the imagination is most evident when discussing alternative economic systems, but according to Haiven, it also influences our identity and values, shaping how we perceive the world and act within it, as well as how we "reproduce our world and are reproduced by it."[9]

WORLDMAKING INTERACTIONS AND ESCAPE VELOCITY

To address this crisis and encourage creative exploration beyond the boundaries of the present, Parisi's concept of "speculation" has emerged as a liberating form of creative inquiry. She suggests that such speculation opens new realms for imagining potential facts, which in turn can unleash the "effective power of the new," allowing us to venture into the realm of the future.[13] According to Bendor, interactive media can afford experiences of various such speculative futures.[14] Such futures are *imaginaries* of sustainability and expressions of the malleability of the future. Through the use of interactive media, Bendor explains, it is possible to achieve a collective reformulation of what is not only possible or probable, but more important *desirable* in the minds of the public, which can allow for an increased or rediscovered ability to "unmake and remake the world."[14] He treats this as a "politico-ontological capacity" of the media, and terms it *Worldmaking Interactions.*[4] WI incorporate a diverse set of artistic techniques, where a blend of playful ambiguity and open-endedness serves to stimulate a collective reimagination of the future.[4,14] Thus, WI aspire to transcend conventional expectations of the future, providing experiences that are distinctly unconventional and divergent from established projections. One of the main objectives of WI is to establish a flow of ideas between individuals' imagination and the prevailing social imaginaries.

Concerning sustainability, Bendor claims that it cannot be approached as a "complex problem to solve" through technological innovation or scientific breakthrough, since this limits the range of futures for which we may strive.[4] Instead, he argues for the potential for interactive media to "discern, unpack and reframe connections between our values, worldviews and beliefs about the very nature of the world, and our capacity to act on it."[4] His understanding of sustainability is close to that of Robinson, who argues that sustainability should be thought of as an "emergent property" of conversations about desired futures that "is informed by some understanding of ecological, social and economic consequences of different courses of action."[15]

Bendor also introduces the notion of "escape velocity."[14] Drawing on Deleuze and Guattari, Bendor defines escape velocity as "envisioning and pursuing … 'lines of flight', generating momentum for radical transformation."[14] We understand escape velocity as a performative potential generated through WI, and one way that can help mitigate the crisis of the imagination. By engaging in WI, individuals or groups of individuals can come to acknowledge the potential for alternative futures, fostering momentum for transformations.[10]

Bendor discusses WI in the broad context of interactive media in general. To illustrate this design philosophy, Bendor et al. set up a multimedia exhibition called *Sustainability in an Imaginary World*.[16] Aspects of WI, such as playful ambiguity and open-endedness, were applied to the installation, for instance through unclear instructions of how to move between the different rooms and through the use of symbols. Participants were thus invited to form their own ideas and project their own imaginations, and were thrown between various contemplative, abstract, and unguided experiences.[4]

WORLDMAKING INTERACTIONS IN VIDEO GAMES

While various forms of interactive media, whether purely digital or a combination of digital and analog, may serve as suitable platforms for fostering WI, one of our main contributions of this paper is to show and argue that the video game medium specifically can be an ideal vessel for such design.[11] In this section we will present a number of arguments supporting this thesis, and present examples of how WI have already been introduced to conventional games.

First, as we have already established, the impact and pervasiveness of the medium on society is hard to overstate. Developers and publishers spend millions of dollars and working hours on crafting immersive narratives and environments, complex characters and engaging mechanics and gameplay loops in order to keep players occupied for hundreds of hours. This type of experience is incomparable to anything else in the realm of digital media at present. The sheer popularity and scale of the video game medium thus make it potentially more impactful than other forms of media,[16] since WI relies on upholding active communities of people who engage with the "unarticulated or 'unfinished'" futures afforded by the games.

Second, the participatory and collaborative nature of gaming makes video games an ideal medium for exploring questions about ourselves and our coexistence with our environment.[17] Games have long had the capacity to evaluate players' moral principles by presenting them with "meaningful" choices, which promotes reflection and an increased emotional and empathic relationship to the game itself, its narrative and game world.[11,17] The step from those reflections towards collective discussions is getting increasingly shorter with the help of online forums such as Reddit and online communication apps such as Discord.

Third, the unique and engaging blend of ludological and narratological elements in video games, which deeply immerses players in their game worlds, stands as a catalyst for WI. This is attributed to the possibility to apply key aspects of WI, such as playfulness, open-endedness and imagination into video games.

Fourth, WI acknowledges the productive role of feelings such as discomfort, uncertainty and ambiguity for interaction design. These kinds of feelings are often evoked in for example art exhibitions and movies, but the role of them in video games is often understated, as games and play are more commonly associated with sensations of fun and enjoyment.[1] Still, these more "negative" but nonetheless productive feelings are also evoked by video games to promote engagement.[2]

As put by Vervoort, games are unique since they allow people to explore new roles, rules and interactions in strongly realized worlds, using a mix of interactive media.[18] Reflecting on Bendor's original conceptualization of WI in interactive media, and how we have expanded on this in the context of video gaming, it becomes evident that some video game developers have already begun incorporating WI into their games.

In *Dark Souls III*, which is set in the crumbling Kingdom of Lothric, the player assumes the role of an Unkindled tasked with rekindling a fading flame. The narrative is unveiled gradually, but never completely, through cryptic item descriptions and atmospheric design, encouraging players to interpret the world. While player actions have consequences for the future of the game's world, the actual futures are never fully articulated, but open for interpretation even after the story has been concluded. Furthermore, the game offers little guidance and invites to playful exploration and trial-and-error. The game is also known for being frustratingly difficult and unforgiving, giving rise to all sorts of feelings and emotions while playing. Furthermore, while the game is mainly a single player experience, it is possible to communicate with other players through in-game messages that appear in other players' game worlds. This fosters a sense of community and shared knowledge among players, even though there is no direct collaboration or communication between them.

While not all video games are designed to promote WI, some unintentionally facilitate these interactions through features like open-endedness and playful ambiguity, which are often included for either entertainment or artistic purposes. These aspects can create opportunities for players to engage with the game's environment in ways that encourage exploration and creativity.

Although we contend that conventional video games have the potential to foster WI for sustainability purposes, the connection between WI and visions of environmentally sustainable futures is not immediately apparent in the current understanding of WI. To steer the futurescaping resulting from WI in that direction, we therefore argue that Environmental Empathy (EE) could be integrated into this interaction design framework.

ENVIRONMENTAL EMPATHY

It is now commonly known that merely providing information about sustainability is an ineffective or at least insufficient way to promote transformative, sustainable change.[11] Scholars and climate activists alike instead argue that fostering empathy is essential for promoting environmental awareness, understood as an emotional state reliant on one's ability to care for the experiences of other persons, animals or elements of the natural world.[7] Blythe et al. argue that empathy for nature is needed in order for sustainable interactions with the biosphere to emerge.[7] According to Ruberg, the focus on fostering empathy through the use of video games has increased in recent years, since the medium is suitable for allowing players to experience the feelings of other, albeit fictional, human and non-human actors.[19] In this context, EE specifically implies fostering a deep connection to the natural world and sense of responsibility towards it.[7]

Blythe et al. explore the effectiveness of using virtual reality (VR) to depict future scenarios focusing on the human impact on oceans, demonstrating its ability to cultivate empathy for nature and the environment. The researchers let respondents experience both an optimistic and a pessimistic scenario, with the pessimistic scenario resulting in significantly greater increases in empathy. This challenges the idea that positive news stories are the primary drivers of support for conservation and stewardship. Still, the study finds that the heightened

empathy experienced by the participants was temporary, as it eventually reverted to baseline levels. Regardless, they conclude that providing impactful depictions of future scenarios have the potential to bridge the "empathy gap" necessary to start caring more for our common future.

While environmental sustainability is clearly a major concern for Bendor, the design principles proposed by the WI framework itself (e.g., open-endedness, ambiguity, playfulness) do not necessarily put emphasis on environmental issues, as shown in the *Dark Souls* example in the previous section of the paper. While realizing that many future sustainable imaginaries resulting from WI might indeed imply increased environmental care and stewardship, this is not necessarily the case. Rather, a desirable future from an anthropocentric perspective may instead entail asserting control and dominance over nature and the environment.

By integrating EE into WI design, games can evoke a greater awareness of environmental issues and the impact of human actions on our surroundings, and ensure that also non-human actors have a "voice" in futurescaping activities. This heightened awareness can lead to more mindful decision-making within the game, promoting sustainable practices and values which can be shared and collectively articulated among players. Moreover, EE can motivate players to apply these experiences outside the game, encouraging real-world behaviors that support environmental sustainability. Furthermore, as will be shown in the empirical results of the paper, EE in-game could also function as a "trigger" to WI.

METHODOLOGY

In the empirical part of this study, which was conducted in the first half of 2023, our aim was to explore how existing video games promote WI and produce escape velocity towards co-produced futures, and the role EE played in these interactions. Therefore, we actively sought respondents who had experienced in-game interactions that connected in-game mechanics or narratives with real-world, sustainability-related issues. Throughout several posts on video game-related forums on the discussion platform Reddit, we asked to be contacted by anyone who had experienced such moments in-game, regardless of whether the game itself had explicit sustainability-related themes or not. Thus, this part of the study was not conducted to see how effective games in general are to accommodate for such interactions and experiences, but to search for example of how WI and EE can potentially co-produce escape velocity to provide inspiration for our theoretical development.

The second author was during the data collection period contacted by 16 respondents (13 M, 3 F, between 14 and 39 years old from North America, Asia and Europe) and conducted interviews with them using the online communication platform Discord. Due to the explorative nature of the study, we found open-ended interviews suitable for achieving this aim, and attempted to approach respondents in diverse cultural and geographical contexts. Despite our efforts to achieve diversity, however, we were unable to achieve a balanced representation of different genders, which can be considered a limitation of the study.

After the interviews were concluded, the audio recordings were transcribed and color-coded in accordance with emerging themes, e.g., *games as a catalyst for change*, *games and symbolism*, *games and building empathy* and *games as a way of reinforcing sustainability*. Responses correlating to these themes were then analyzed through the lens of our conceptual framework.

RESULTS: IN-GAME INTERACTIONS AND REAL-WORLD SUSTAINABILITY IMPACTS

In our discussions with gamers, it became immediately evident that the video game medium is distinctly positioned to deliver immersive and captivating experiences that surpass the scope of traditional media. Several respondents expressed that it can afford new ways of dealing with real-world issues and produce new narratives through imagined or symbolic versions of real-life problems. More interestingly, some of the interviewees provided examples of how their involvement with specific games had not only increased their awareness of various social and environmental issues, but had also transformed their real-life behaviors.[14] First, we would like to present some examples of how these interactions can materialize in practice.

One of our interviewees (R3) discussed *Final Fantasy VII*, a game whose storyline has profoundly impacted them since their childhood. It revolves around a powerful corporation exploiting the planet, sapping its life energy to fuel a sprawling mega city. While the game can be interpreted as having an environmental message, the respondent only started to reflect upon these themes later in their life, drawing parallels between the narrative and the unethical business practices of oil companies and the effect of fossil fuels on the global climate. The interviewee attributed certain decisions in life, such as not buying a car with a combustion engine, to their interactions with the

game years prior, which sparked a deeper care for the planet.

Another interviewee (R7) presented a similar experience within the game *Horizon: Zero Dawn,* which is an action RPG set in a post-apocalyptic North America dominated by mysterious robotic creatures. The interviewee was employed by an NGO at the time, and describes themselves as pessimistic and lacked positive visions about the future, believing that despite their best efforts, their actions had no discernible positive impact. After playing the game, they came to the realization that the small things also mattered and were worth fighting for, even in the big scheme of things. As they delved into the universe of *Horizon: Zero Dawn* as the main protagonist, Aloy, they cultivated an in-game mantra, "It's all about hope." Despite confronting seemingly insurmountable challenges in the endeavor to rescue a game-world scarred by humanity's past mistakes, they managed to endure and kindle hope among the diverse settlements within the game. The resilience of the protagonist resonated with the respondent, reshaping their perspective also of the outside world. They learned to embrace the idea that our limitations do not render us powerless, but instead serve as points of recognition to help us transcend them. This insight prompted them to embody their in-game motto in real life, particularly in their professional role but also in their personal life, where they for instance decided to adopt a vegetarian diet.

The *Horizon* franchise was discussed frequently by several respondents, not least since the worldbuilding within those games manage to communicate relatable futures. They are also abundant in symbolism, referring to real-life people, places and events that give rise to a plethora of online theories and discussions. There are, for example, those who claim that many of the named NPCs responsible for the in-game apocalypse represent real-life leaders and tech gurus. One of the interviewees (R4) drew a connection between the in-game Red Blight in *Horizon: Forbidden West*, which affects over-fertilized soils, and real-world sustainability challenges like over-fertilization, acidification, and climate change which, in the near future, can threaten the world's food supply. Another respondent (R2) emphasized that the environmental problems faced by the game's tribes amplified their capacity to empathize with real-world marginalized and impoverished communities most severely affected by global and local environmental issues. Another respondent (R10) came to similar conclusions after playing *Ghost of Tsushima*, a title set in a fictional version of Japan where the main protagonist has to protect his homeland from Mongol invaders. Living in a country with high levels of corruption and an authoritarian regime, the respondent found a deep sense of empathy with this struggle for democracy and sustainability in a real-life context.

Yet another participant (R6) shared their experience of empathizing with the characters and the world portrayed in the *Horizon* franchise, evoked by the in-game challenges related to climate change, water resources, and agriculture. The respondent noted that this empathy stemmed from recognizing the striking parallels between the issues plaguing the *Horizon* game world and those confronting our real world. Another game that was frequently brought up was *Cyberpunk 2077*, which has in common with the *Horizon* franchise that it is depicting a dystopian future where ultra-modern technology coexists with a degenerate human society, and where players are immersed in the universe through forcing them to navigate a hyper-capitalist world where everything is for sale.

ANALYSIS: WORLDMAKING INTERACTIONS AND ENVIRONMENTAL EMPATHY IN CONVENTIONAL VIDEO GAMES

In the previous section, our aim was to provide some examples of how video game design can promote interactions between the player and the game-world that exerted performative impacts on their attitudes towards sustainability in real-world settings. In this section, we aim to show how such interactions are indeed influenced by the main components of WI design, such as *open-endedness*, *playfulness*, *ambiguity* and *discomfort* as well as EE, more specifically *emotional responses and connections*.[7,14]

On a general level, one of the primary insights gleaned from results is the notable impact of conventional games on individuals' conceptions about and perspectives on sustainability, as well as their abilities to interpret, assimilate, and produce future visions and scenarios, regardless of whether the game itself contained explicit sustainability-related messages. The respondents instead independently constructed their own interpretations of the narratives and mechanics presented in the games and let these insights influence their perception of environmental sustainability, but also social sustainability issues such as justice and democracy.

Our dialogues with gamers especially underscored the efficacy of *environmental storytelling*, a narrative

approach that does not follow the traditional method of conveying a coherent story through for example cutscenes. Instead, the story gradually unfolds in an unstructured manner, offering – sometimes conflicting – narrative fragments piece by piece by visual, textual and auditory means. This kind of storytelling can promote *ambiguity* and *uncertainly* in the interaction with the game-world, since the information about the game's world is left intentionally vague and incomplete.

While an individual's actions towards and visions of sustainability are interesting and important to take into consideration, WI hinges on the fact that visions and experiences are collectively interpreted, shaped and shared. Especially for the *Horizon* franchise discussed by many of the respondents, the ambiguity and uncertainty afforded by environmental storytelling have sparked extensive online discussions about the game's lore and real-world inspirations and implications on various forums and communication apps.[1] In this context, *playfulness* plays an important role in crafting theories about the game itself and asking important questions about the game's messages and implications for future visions and imaginaries. This may give some insight into one of the questions asked by Vervoort et al., namely "how gaming communities and cultures contribute to the development of different future imaginaries that are relevant to sustainability concern."[8]

As we have seen, video games and other interactive media can inspire new visions and imaginaries by providing players with examples of alternative social, political, economic, or cultural configurations.[14] In the games discussed by the participants, viable and desirable visions and imaginaries are never explicitly stated or suggested; instead, the focus is on visualizing dystopian versions of a future that are to some extent probable given how real-life societies are organized today. The struggles of the protagonists often revolve around navigating the negative effects of continued technological development, resource exploitation and environmental destruction. In line with Blythe et al., it seems as if these dystopian futures have a potent ability to trigger *emotional responses and connections* with in-game human and non-human actors, showing the potential of such games to induce EE.[7] More importantly, however, they are able to translate these empathic feelings into a real-life context, also concerning aspects related to sustainability. Bendor states that the semiotic elements within interactive media enable users to connect with the characters and settings, effectively mediating the world portrayed.[16] Looking at the discussions concerning those games online (see the footnote above), and from listening to our respondents' responses to those in-game interactions, it seems as if EE can act as a trigger to achieve escape velocity, understood as a collective generation of momentum for radical change.[14] This momentum is not only reliant on positive visions of the future, but seem to be triggered by emotions such as *discomfort* or *despair* resulting from being immersed in a fictional game-world while also being able to "connect the dots" to various trajectories, developments or events in the real world.[16]

CONCLUSIONS

In this paper, we have argued that WI, as a theory and design philosophy, should be translated into a video game design context for environmental sustainability purposes. We have provided theoretical and empirical support for design that promotes WI in video games, and recommend incorporating elements that evoke EE to maintain a clear emphasis on environmental sustainability, driving the momentum for radical change (escape velocity) that can arise from these interactions. This endeavor was motivated by the increased pervasiveness of video games in contemporary societies. Furthermore, in Sweden and other Western contexts, the number of young people identifying as *gamers* has increased steadily since 2019,[2] while more progressive identities such as *environmentalist* are on the decline. While we hesitate to say that video game developers *must* promote sustainability, we find it vital to contribute to academic discussions on sustainability and game design, since this may provide developers with the tools necessary to raise awareness and encourage action.

Previous conceptualizations showcase how a mix of digital and analog media can be used successfully to provoke WI for speculative futurescaping. What we have managed to show is that the video game medium can be a perfectly competent vessel to promote these kinds of interactions on its own. We have shown how *playful ambiguity*, *open-endedness* and *symbolism* are used to great extent in such games to invite to theorizing and co-production of futures, to move from the level of the individual to the collective.[4] Moreover, because video games attract a large global audience, they are potentially more "democratic" than events limited to specific physical locations, such as exhibitions.[16]

[1] See e.g., https://tinyurl.com/hzancients, https://tinyurl.com/hzmats, https://tinyurl.com/hzbact.

[2] According the *Swedish Youth Barometer*, https://www.ungdomsbarometern.se/rapportslapp-generationsrapporten-2024/.

However, while environmental sustainability can be considered a central component to WI design, we worry that the lack of explicit environmental sustainability components in the framework may give rise to visions that do not take these aspects into account. Such visions could very well involve increased exploitation of natural resources, such as mining the ocean floor for rare minerals, deforestation for agriculture and urban development, or colonization of space to ensure the survival and expansion of our species beyond Earth itself. Geo-engineering is another example. Such purely anthropocentric futures would overlook inherent values of and spell disaster for non-human actors.

This insight led us to explore the potential of explicitly including aspects of EE to our reconceptualization of WI design for video games. However, throughout our interviews, we also realized that emotional responses and connections acted as catalysts for escape velocity in WI design. In a study by Blythe et al., they came to the conclusion that non-interactive VR experiences designed with EE in mind heightened the empathy levels among their respondents only temporarily.[7] However, our respondents reported long-lasting effects from similar experiences in conventional video games. This suggests that the interactive nature of the medium, which treats the player as an actor rather than a spectator, can be more impactful.

There is critique towards the recent rise of so-called "empathy games." Ruberg, for example, suggests that efforts to foster empathy through oversimplified narratives or characters can inadvertently obscure the intended emotional response.[19] Therefore, she argues, it may be more effective to promote empathy for those we do not or cannot currently understand. This claim is in line with our results, that suggest that video games can promote empathy for non-human actors, including plants, animals, natural environments, and the climate.

According to Vervoort et al., there is still an emphasis on the role of serious or educational games to promote sustainability-related impacts.[8] Our interest in conventional games arises from the fact that none of our respondents reported WI or emotional responses and connections in games designed for purposes other than entertainment. This is not to say that such interactions are impossible in these games, but rather that serious or educational games remain a relatively niche phenomenon. While there is some merit to the claim that we should start breaking down the walls between serious and conventional games,[8] drawing on our reconceptualized WI framework, we also see some merit to keeping sustainability-related themes vague, and depict "desirable futures" in an unfinished and open-ended state to promote speculative futurescaping. Ultimately, most people engage with video games for the enjoyment they offer, and focusing too much on their potential positive "side-effects" could be detrimental, especially for those not initially interested in environmental sustainability.

In conclusion, WI, and the escape velocity generated by such interactions, does not automatically lead to transformative change through for example political action. Nevertheless, the ability to acknowledge the profound unpredictability of the world and to conceive of alternative possibilities serves as the foundation for societal transformation.[10] While this in itself might not be sufficient, it is unquestionably essential for forging new visions and, ultimately, driving change.

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